Microsoft

Reconstructing Group Cognition for Collaborative Agents

Zhaoxuan Tan[1*], Mengting Wan[2], Jingbo Yang[3], Kwei-Herng Lai[2], Renyi Qu[2], Lorenzo Jaime Flores[2], Meng Jiang[1], Longqi Yang[2], Victor Poznanski[2] and Pei Zhou[2]

[1]University of Notre Dame, [2]Microsoft, [3]UC Santa Barbara

Abstract

Agents entering an ongoing collaboration need context about how the team thinks and works: which beliefs participants have expressed, which responsibilities have been accepted, and where coordination may break down. We introduce *group briefing*, the task of reconstructing this task-relevant group cognition from a collaboration request and the interaction history available at a specified cutoff. We build a synthetic benchmark with 1,351 requests and 9,975 verified reference points across four domains, and evaluate both information coverage and downstream collaboration. Comparisons of full-context prompting, retrieval, persistent memory, and our task-guided retrieval and synthesis method, GroupBrief, reveal a consistent challenge: recovering past state is easier than identifying future friction. Across nine writers using GroupBrief, full-benchmark coverage ranges from 14.56% to 53.24%. We then study whether coverage feedback can improve briefing policies with the writer fixed. The resulting GroupBrief (optimized) raises Test coverage by 2.54 percentage points for `gpt-5.4-mini` and 12.88 for `gpt-5.6-luna`; their simulation preference rates against no brief are 64.46% and 68.42% on valid Test comparisons. Together, the task, benchmark, and analyses provide a basis for studying how agents turn distributed interaction history into useful collaboration context.

Project Page: https://zhaoxuan.info/groupbrief/

Correspondence: ztan3@nd.edu, pei.zhou@microsoft.com

1. Introduction

An agent asked to support an ongoing team must understand how its members' knowledge, responsibilities, and plans fit together. A project handoff may depend on one member recognizing a revised deadline, another resolving an approval dependency, and both knowing who will act next. Such relationships are distributed across messages and evolve as participants propose, revise, and acknowledge decisions. Research on common ground, mutual knowledge, and coordination explains why these relationships matter to collective work (Clark and Brennan, 1991; Cramton, 2001; Okhuysen and Bechky, 2009). For an agent entering the team, they create a *collaboration cold-start problem*: access to the history does not by itself provide an account of the group's current coordination needs.

Existing research supplies important parts of this preparation. Long-term memory benchmarks examine recall, temporal reasoning, and changing information (Maharana et al., 2024; Wu et al., 2025); multi-party benchmarks study attribution and evolving participant-specific context (Hu et al., 2026a; Owolabi, 2026; Yang et al., 2026); and social-agent evaluation studies goal-directed interaction (Zhou et al., 2024). These capabilities raise a further question: *what should an agent recover from a group's history before participating in a particular collaboration?* In Figure 1, the agent needs to connect a revised handoff date to a teammate's outdated plan, recognize a pending approval, and identify how a shared checklist could help. The useful context lies in the relationships among these facts and their implications for the requested work.

*Work done while interning at Microsoft.

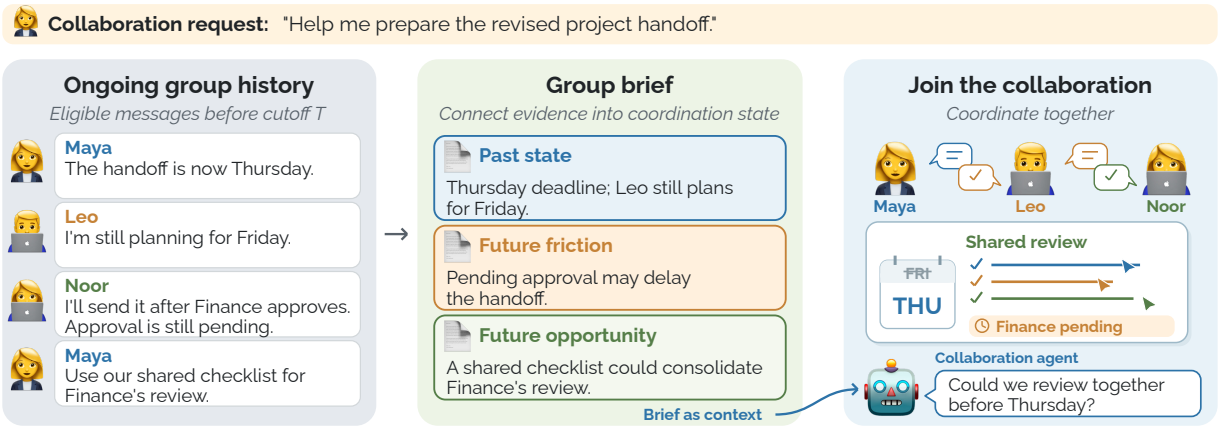


Figure 1 | Group briefing addresses a collaboration cold start. In this illustrative handoff, the agent must connect scattered messages to recognize a deadline mismatch, a pending dependency, and an opportunity to coordinate before joining the team's work.

We introduce *group briefing* to study this preparation. Given a collaboration request and a cutoff time, a briefing agent constructs an evidence-grounded account of task-relevant *group cognition*: participants' expressed beliefs, group dynamics, and gaps in shared understanding. The brief covers past team state, evidence-supported future friction, and opportunities for collaboration. It preserves distinctions that affect coordination: an offer to help is different from accepted ownership, and a posted update is different from acknowledged understanding. The resulting brief supplies context to a collaboration agent, which can use it to decide what to clarify and which dependencies to resolve. The briefing agent itself neither answers the request nor makes commitments for the team.

To evaluate group briefing, we construct a benchmark of 1,351 collaboration requests and 9,975 reference points from synthetic interactions in manufacturing, finance, healthcare, and technology. Model-assisted generation and verification identify the historical and prospective information relevant to each request. Prospective points require pre-cutoff support and corroboration in later interaction; later evidence is used for annotation and excluded from runtime inputs and the coverage judge's scoring view. We evaluate two complementary aspects of a brief. Reference-point coverage measures the coordination information it retains. Collaboration simulations measure preference between interactions produced by a fixed coordinator with and without the brief under matched scenarios and action budgets.

Our experiments compare full-context prompting, retrieval methods, persistent memory systems, and GroupBrief, which iteratively gathers task-relevant evidence and synthesizes a group brief. Two findings stand out. First, recovering prospective coordination difficulties remains challenging: across nine writers using GroupBrief, full-benchmark coverage ranges from 14.56% to 53.24%, and every writer covers past state better than future friction. Second, holding the writer fixed still leaves a wide range of coverage across briefing methods. With `gpt-5.4-mini` fixed, GroupBrief reaches 32.54%, compared with 31.95% for Hindsight (Latimer et al., 2025) and 14.48–26.91% for the other baselines. These are comparisons of complete configurations with different context and memory-construction budgets. They expose substantial missing coordination information even in the highest-coverage briefs.

These findings motivate a further study of coverage-guided policy optimization. Keeping the writer fixed, we revise GroupBrief's retrieval and synthesis instructions through five rounds of proposals, training feedback, and validation selection. On 289 Test requests, the resulting GroupBrief (optimized) increases coverage by 2.54 percentage points for `gpt-5.4-mini` and 12.88 for `gpt-5.6-luna`. Their optimized briefs obtain tie-adjusted simulation preference rates of 64.46% and 68.42% against no brief on each configuration's valid comparisons. A case study illustrates what the revised instructions emphasize: following decisions through later pre-cutoff updates, separating the person who confirms an arrangement from its responsible owner, and retaining the

arrangement's conditions and purpose. Optimization thus provides a concrete way to investigate how briefing policies can address the information gaps exposed by the benchmark.

Our contributions are threefold. First, we formulate group briefing as the reconstruction of task-relevant group cognition for agents entering ongoing collaboration. Second, we build a benchmark with complementary coverage and interaction evaluations and characterize the challenges faced by memory methods and writer models. Third, we introduce GROUPBRIEF and study coverage-guided policy optimization, connecting aggregate results to the coordination information preserved in individual briefs.

2. Method

Group briefing prepares an agent for ongoing collaboration by reconstructing task-relevant group cognition: participants' expressed knowledge, interdependent responsibilities and plans, and unresolved gaps in shared understanding. We formulate the task and benchmark, introduce GROUPBRIEF, and evaluate briefs and subsequent interaction (Figure 2). Section 5 separately studies coverage-guided policy optimization.

2.1. Group Briefing Task

Let g denote a team and $\mathcal{H}_g$ its interaction history. A task instance is $x = (g, q, T, \mathcal{H}_g^{\leq T})$, where q is a teammate's collaboration request, T is the cutoff time, and $\mathcal{H}_g^{\leq T}$ contains messages permitted by the task's visibility rules. Each message includes its identifier, author, timestamp, content, and available reply relationships. The briefing agent receives the public team and project scope and accesses only eligible messages at or before T. Reference annotations, generator metadata, hidden evaluation conditions, and later messages remain outside this input boundary.

For a fixed language model M_θ and briefing policy π, the brief $b_x = F_{\theta,\pi}(x)$ prepares another agent for the requested collaboration. The brief connects *past state*, such as expressed positions, decisions, and responsibilities, with *future friction*, or supported concerns that may obstruct the task, and *future opportunities*, or supported ways to improve coordination. It preserves who said or accepted what, when it held, and what remains contingent. For example, an offer to help does not establish accepted ownership, and an update does not establish acknowledged understanding. It supplies context without answering the request or making commitments for the team.

2.2. Benchmark Construction

Interaction histories and requests. We construct synthetic group histories in manufacturing, finance, healthcare, and technology following GroupMemBench (Yang et al., 2026). To maintain continuity across exchanges, generation uses participant personas, recent messages, and a running ledger of decisions and commitments. A typed graph connects projects, topics, phases, participants, and messages through membership, authorship, and reply relations. A planning stage selects speakers and intents, a writing stage generates each exchange, and speaker-specific rewriting differentiates voices. Validated messages receive phase-based timestamps and update the graph and ledger. From each inherited visibility snapshot, `gpt-5.5` generates requests about upcoming collaborative activities while withholding the facts needed to address them.

Reference points with temporal boundaries. To specify what a useful brief should preserve, we pair each request with task-relevant reference points G_x. The request-generation call proposes past-state candidates; friction and opportunity points are mined separately. Each point identifies a work object, actors or roles, cutoff status, *core facets* needed to convey its meaning, and *supporting facets* that add detail. Past-state points describe established arrangements and expressed team states. Prospective points must link a pre-cutoff signal to a later realization of the same obstacle

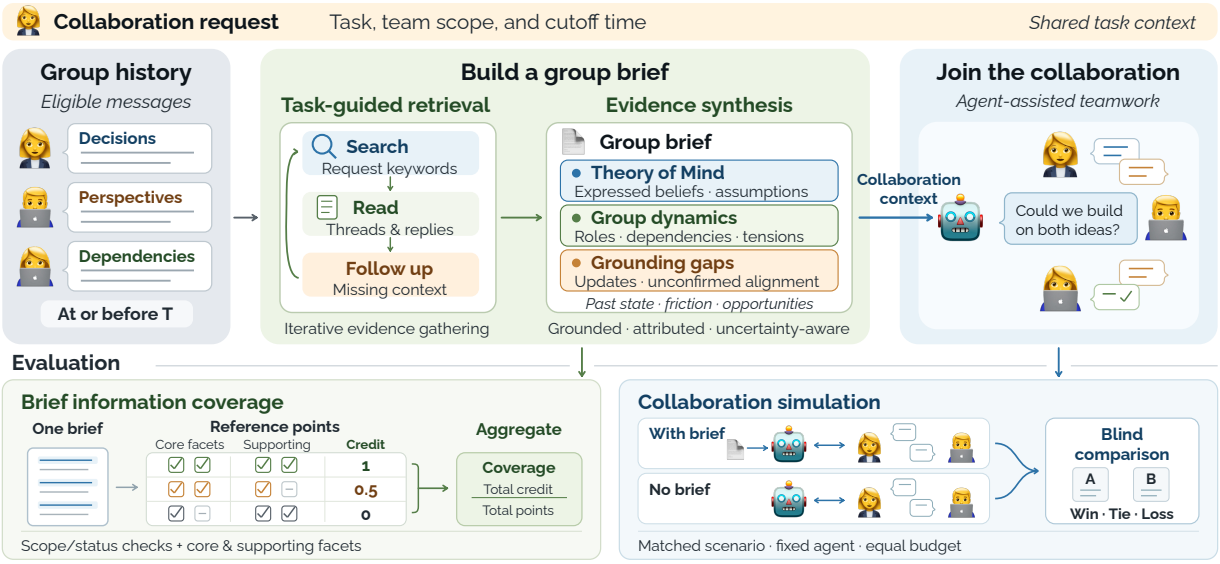


Figure 2 | Overview of group briefing and evaluation. A collaboration request guides iterative retrieval from eligible group history. Evidence synthesis connects participants' expressed beliefs, group dynamics, and grounding gaps into a group brief. The brief supplies collaboration context to an agent joining the team. Evaluation measures reference-point coverage and compares interactions with and without the brief under matched conditions.

or intended benefit. Later interaction validates the point's inclusion, but its scored assertion and facets concern only what was identifiable at T. We store later evidence separately and exclude it from both the briefing agent and the coverage judge.

Verification and release. We verify candidate points through separate model calls that check historical support, temporal validity, task relevance, atomicity, category fit, and consistency with generator metadata. Deduplication and category caps limit redundancy. Each released request retains at least five points across at least two of the three reporting groups, yielding a frozen benchmark $\mathcal{D} = \{(x, G_x)\}$ with 1,351 requests and 9,975 points. The points are task-specific propositions rather than unique source messages: multiple requests may draw on the same history. Appendix A details generation windows, verification, and release selection.

2.3. GroupBrief

GroupBrief uses two stages to turn a collaboration request into usable context: task-guided evidence acquisition and brief synthesis. Its policy $\pi = (I_{\text{ret}}, I_{\text{syn}}, \kappa)$ comprises retrieval instructions I_{ret}, synthesis instructions I_{syn}, and control logic κ governing evidence organization, budgets, and stopping. The policy is fixed across instances within a writer-specific experiment. The evidence gathered under that policy adapts to each request and the observations returned during retrieval.

2.3.1. Task-Guided Evidence Retrieval

Task-guided retrieval aims to recover the relationships needed for the requested collaboration. Relationships can span exchanges, so retrieval proceeds iteratively: a message naming an owner can prompt inspection of that person's response, and a schedule change can prompt a check of later acknowledgments. At turn t, the model selects an action from the task context and retrieval transcript τ_t:

$$a_t \sim M_\theta(\cdot \mid \Phi_{\text{ret}}(x, \tau_t; I_{\text{ret}}, \kappa)), \tag{1}$$

where Φ_{ret} constructs the retrieval prompt. An action may issue one or more tool calls for lexical search, participant lookup, or message access by author, thread, reply chain, mention, or time window. All tools operate on $\mathcal{H}_g^{\leq T}$; their results extend τ_t and inform the next retrieval decision.

Retrieval starts with an empty transcript and requires at least two tool calls before finalization. It stops when the model finalizes or exhausts its turn budget. We assemble the identified messages into an evidence set E_x, preserving source identifiers, authors, timestamps, and content for synthesis. If retrieval identifies no messages, a fallback lexical search uses the request and task scope. Appendix B specifies the tool interface and budgets.

2.3.2. Synthesizing Group Cognition

The synthesis stage connects the collected evidence into a coherent account of the team's collaboration state. A separate model call generates a draft brief:

$$\widetilde{b}_x \sim M_\theta(\cdot \mid \Phi_{\text{syn}}(x, E_x; I_{\text{syn}}, \kappa)), \tag{2}$$

where Φ_{syn} assembles the synthesis instructions and evidence. The request shapes evidence selection, while GroupBrief's synthesis prompt contains the scope, cutoff, and collected messages without repeating the request text.

We organize synthesis around three complementary dimensions of group cognition. *Theory of mind* captures participants' expressed beliefs and assumptions, preserving attribution and uncertainty. *Group dynamics* captures roles, accepted responsibilities, dependencies, and tensions around shared work. *Grounding gaps* identify updates or arrangements whose acknowledgment or alignment remains unconfirmed; absence of acknowledgment alone establishes neither ignorance nor agreement. Together, these dimensions guide the writer to connect proposals with acceptance, revisions with earlier plans, and responsibilities with their prerequisites. Friction and opportunities are conditional implications supported at the cutoff.

The resulting brief uses six headings: Collaboration State, Group Dynamics, Disagreements, Theory of Mind, Grounding Gaps, and Lookahead. These headings structure the prose; the method requires neither a separate mental-state estimator nor a graph representation. Evaluation considers the entire brief and does not require this output structure. To produce b_x, a final check removes citations to message identifiers outside E_x. This check validates identifier membership; it does not determine whether the cited message supports the claim.

2.3.3. Brief as Collaboration Context

A fixed collaboration agent receives the brief, request, and other permitted context. Acting as coordinator, it uses this context to guide clarification, contacts, and prerequisite resolution, with access to further retrieval and member communication. The brief is private to the coordinator: members' understanding changes only through information communicated during interaction. This boundary, fixed agent, and fixed action budget let us evaluate the preparation supplied by the brief.

2.4. Evaluation Framework

We ask whether a brief retains required coordination information and helps a coordinator in subsequent interaction. Reference-point coverage measures information retention and supplies policy-optimization feedback. Collaboration simulation assesses interaction under matched with-brief and no-brief conditions.

2.4.1. Reference-Point Coverage

To measure information retention, an LLM judge checks each reference point $p \in G_x$ against the brief. It assesses the point's object, actors or roles, cutoff status, and core and supporting facets, labeling each component as supported, missing, or contradicted. Supported and contradicted judgments must cite brief lines. The judge sees only the semantic checklist and brief, excluding the source conversation, generator metadata, and later evidence. Reference points serve offline evaluation and feedback and are never supplied to the briefing agent during generation.

We convert these component judgments to point credit with a deterministic rule, $s(b_x, p) \in \{0, 0.5, 1\}$. A point receives zero if its object or temporal status is missing, any scope component is contradicted, any core facet lacks support, or any supporting facet is contradicted. Otherwise, missing supporting facets yield 0.5 and complete support yields 1. Missing actor scope alone does not force zero, although an actor required by a core facet must be covered. We also check that cited line identifiers belong to the evaluated brief.

For $\mathcal{D} = \{(x_i, G_i)\}_{i=1}^{n}$ and generated briefs $b_i^{\pi} = F_{\theta,\pi}(x_i)$, coverage aggregates credit over all reference points:

$$J(\pi; \mathcal{D}) = \frac{1}{N_{\text{ref}}} \sum_{i=1}^{n} \sum_{p \in G_i} s(b_i^{\pi}, p)\,, \tag{3}$$

where $N_{\text{ref}} = \sum_{i=1}^{n} |G_i|$. We report $100J$, weighting points equally across requests. This measures retention of the reference information; it does not separately score factual precision, unsupported additional claims, brevity, or cost.

2.4.2. Downstream Collaboration Simulation

To assess downstream usefulness, we compare the same coordinator with and without a brief in a matched collaboration scenario. Each scenario specifies eligible history, active members and roles, the request, and a public follow-up. We hold the scenario, initial member information, other context, and action budgets fixed. The coordinator may search history, open messages, contact members, or finish. Members respond using their own authored history and messages actually delivered to them, preserving differences in participants' information.

An LLM judge assesses the resulting collaboration from anonymized interaction trajectories and members' initial and final reports. Hidden task constraints and progress criteria, derived from verified reference points, guide its assessment of evidence fidelity, participant uptake, and compatibility of intended actions. The judge sees no method names, briefs, or coverage scores. We obtain two judgments with reversed display order (AB/BA) and stable anonymous identities, retaining a win, loss, or tie only when both judgments are valid and agree. Simulation failures, judge failures, abstentions, and order-sensitive disagreements remain recorded exclusions. The tie-adjusted preference rate is

$$\text{WinRate} = \frac{N_{\text{win}} + 0.5 N_{\text{tie}}}{N_{\text{win}} + N_{\text{loss}} + N_{\text{tie}}}, \tag{4}$$

where the counts indicate whether the brief-conditioned coordinator is preferred, dispreferred, or tied with the no-brief control. The rate measures preference conditional on evaluable interactions rather than unconditional task success. Both evaluations draw on the same reference information, so they provide complementary views rather than independent evidence of real-world effectiveness. Appendix C details information boundaries, judging, and exclusions.

3. Experimental Setup

3.1. Dataset

We use the frozen benchmark from Section 2.2: 1,351 requests and 9,975 verified reference points across four domains. The reference set contains 4,576 past-state points, 2,589 future-friction points, and 2,810 future-opportunity points. All methods share the task records and reference sets and follow the same information boundaries. Table 3 in Appendix A gives the domain breakdown. Section 5 introduces the project-level splits used for policy search.

3.2. Models and Comparisons

We examine writer capability and briefing methods separately. First, we evaluate nine writers with GroupBrief on all 1,351 requests and 9,975 reference points: `gpt-5.6-sol`, `gpt-5.5`, `gpt-6-astra`, `gpt-5.6-luna`, `gpt-6-luna`, `gpt-6-sol`, `gpt-5.4-mini`, `qwen3.5-9b`, and `qwen3.8-27b`. Each writer also undergoes single-seed simulation. Then, fixing the writer to `gpt-5.4-mini`, we compare GroupBrief with Full context (Li et al., 2024); three five-chunk retrievers, BM25 (Robertson and Zaragoza, 2009), Contriever (Izacard et al., 2022), and DRAGON+ (Lin et al., 2023); and ReadAgent-S (Lee et al., 2024), A-MEM (Xu et al., 2025), Hindsight (Latimer et al., 2025), HippoRAG 2 (Jiménez Gutiérrez et al., 2025), and Mem0 (Chhikara et al., 2025). Section 5 separately studies coverage-guided policy optimization.

These comparisons share eligible project history and the coverage judge, while allowing each method its implemented retrieval, context, model-call, and offline-memory budgets. Baseline synthesis repeats the request; GroupBrief's synthesis prompt does not. The results therefore compare complete configurations rather than isolate retrieval mechanisms under equal computation. Hindsight's aggregate coverage uses a different brief sample from its subtype analysis and simulation, with identical task records. Appendix D describes each implementation and its resource scope.

3.3. Evaluation Settings

We use `gpt-5.6-sol` for coverage and pairwise simulation judging, and `gpt-5.4-mini` for the coordinator and members. The coordinator retains search and reading and receives four action steps, including at most three contact actions. Member evidence packets target 4,096 tokens but retain required evidence on overflow; this packet limit does not restrict the coordinator's retrieval archive. Final member reports may be reused when received information is unchanged. All 1,351 tasks have verified scenarios and use replicate 0. Each trajectory pair receives two direct AB/BA judgments, without audit-model or tiebreaker calls.

We apply the same bounded retry policy to all simulations. A failed trajectory receives at most two additional whole-trajectory attempts, and we retain the first complete trajectory regardless of its later judgment. Completed trajectories, disagreements, and abstentions are not resampled. Exact-request transport retries can recover failed judge calls while preserving the successful opposite-order verdict. Methods share each task's initial private member reports, pre-interaction scenario, and selected no-brief trajectory. Compatible completed results may be reused across execution batches. Appendix C provides the retry and information-boundary details.

We report baseline and writer comparisons over the full benchmark, and final comparisons involving GroupBrief (optimized) on the 289 Test tasks from seven projects, separately from Train/Valid policy selection. Simulation preference is conditional on order-consistent win/loss/tie outcomes. Each rate uses that configuration's valid comparisons against the shared no-brief control; differences between these rates are not paired method comparisons. Appendix C specifies exclusions and recovery procedures.

4. Results and Analysis

4.1. Main Comparison

Higher coverage does not imply the highest downstream preference. With the writer fixed, GroupBrief reaches 32.54% coverage, followed by Hindsight at 31.95% and other baselines at 14.48–26.91% (Table 1). The 0.59 percentage point (pp) lead over the strongest published baseline describes configurations with unequal context and memory-construction budgets. Simulation ordering differs: Hindsight obtains 64.83% preference against no brief, versus 62.21% for GroupBrief. Information retention and interaction preference thus provide complementary views. Each rate uses

Table 1 | Full-benchmark method comparison with `gpt-5.4-mini`. Coverage and conditional simulation preference against no brief are percentages. Bold marks the highest coverage.

Method	Coverage (%)				Simulation (%)
	Overall	Past state	Friction	Opportunities	Preference
Full context	24.94	34.80	13.36	19.54	59.04
BM25	19.20	26.79	9.85	15.46	58.83
Contriever	18.71	27.14	8.17	14.68	59.74
DRAGON+	18.45	26.01	8.83	15.02	58.51
Mem0	14.48	21.15	7.03	10.46	60.29
A-MEM	19.28	27.40	9.12	15.41	64.02
HippoRAG 2	18.23	27.40	7.44	13.22	60.56
Hindsight	31.95	44.69	**16.72**	25.23	64.83
ReadAgent-S	26.91	40.46	11.39	19.13	62.45
GroupBrief	**32.54**	**45.25**	16.57	**26.57**	62.21

its own valid subset against the shared no-brief control; their difference is not a paired method comparison.

4.2. Performance Across Writers

Table 2 | Full-benchmark writer comparison with GroupBrief. Coverage and conditional simulation preference against no brief are percentages. Bold marks the highest coverage.

Writer	Coverage (%)				Simulation (%)
	Overall	Past state	Friction	Opportunities	Preference
`gpt-5.4-mini`	32.54	45.25	16.57	26.57	62.21
`gpt-5.5`	52.83	**67.81**	33.95	45.84	65.77
`gpt-5.6-luna`	38.82	53.25	20.51	32.19	65.00
`gpt-5.6-sol`	**53.24**	67.48	34.94	**46.92**	67.71
`gpt-6-luna`	36.53	50.48	20.53	28.56	65.61
`gpt-6-sol`	35.19	46.26	23.39	28.02	60.62
`gpt-6-astra`	52.13	67.73	34.72	42.76	66.68
`qwen3.5-9b`	14.56	22.63	6.93	8.43	58.52
`qwen3.8-27b`	50.89	63.55	**35.17**	44.75	63.19

Writer choice changes coverage, but future friction remains difficult. Overall coverage ranges from 14.56% to 53.24% across nine GroupBrief writers, yet every writer covers future friction less well than past state (Table 2). Even `qwen3.8-27b`, the best writer on friction, reaches only 35.17%; `gpt-5.6-sol` leads overall at 53.24%. Simulation preference ranges from 58.52% to 67.71% under the same replicate and execution policy. These rates describe valid interactions, with writer-specific exclusions, rather than unconditional task success. Appendix F.1 specifies valid-outcome accounting and retries.

4.3. Evaluation Reliability

Coverage evaluation completes for every task and reference point without judge errors. The comparison does not isolate the contributions of context truncation, brief length, or retrieval quality.

Test simulations retain 232–256 valid comparisons per configuration out of 289 planned (Appendix F.5). Retaining the first complete trajectory within at most two additional attempts measures performance under recovery rather than first-attempt robustness. Failed attempts and excluded outcomes remain available for audit.

5. Enhancing Group Briefing through Harness Optimization

The benchmark identifies a practical question: can an agent recover more coordination information by changing how it gathers and synthesizes evidence, while keeping its writer fixed? We study this through coverage-guided optimization of GroupBrief's executable policy, or harness. The resulting variant is denoted GroupBrief (optimized); GroupBrief continues to refer to the original method.

5.1. Coverage-Guided Policy Search

Our objective is to improve the information retained in the brief. We use coverage $J(\pi; \mathcal{D})$ from Eq. 3 as the search objective and point-level judgments as diagnostic feedback. Missing or partially covered points indicate where retrieval instructions, synthesis instructions, or brief organization may need revision. Downstream simulation assesses the selected policy separately and supplies no search feedback.

Starting from the original GroupBrief policy π_0, an LLM proposer generates a batch of candidates in each round. It receives the incumbent policy, point-level training diagnostics with judge explanations, and accumulated search history and aggregate scores. Candidates may change retrieval instructions, synthesis instructions, and admissible control logic. Writer parameters, tool primitives, visibility rules, citation cleaning, and scoring remain fixed, restricting the search to the procedure that turns eligible history into a brief.

We select policies by validation coverage after checking feasibility and minimum performance. A candidate must satisfy the interface and budget constraints and complete its training evaluation. It advances to Valid if its Train coverage is no more than $\epsilon = 0.02$ (two percentage points) below GroupBrief; Valid applies the same tolerance relative to GroupBrief on that split. Among GroupBrief and all candidates passing these checks through round r, we retain the policy with the highest Valid coverage:

$$\widehat{\pi}_r \in \arg\max_{\pi \in \mathcal{V}_r} J(\pi; \mathcal{D}_{\text{val}}), \tag{5}$$

where $\mathcal{V}_r$ contains the eligible policies with completed validation. Earlier policies remain eligible, allowing the incumbent to persist across rounds. The proposer receives point-level Train diagnostics and aggregate Valid feedback; Test data and downstream simulations are excluded from proposal generation and selection. This is a finite search over policy programs, without gradient updates to the writer or judge.

After selection, GroupBrief (optimized) applies the frozen policy to new histories. Its tool choices adapt to each request and retrieval transcript, while the policy itself remains fixed. Inference therefore requires neither reference points nor the proposer, and performs no per-team reoptimization.

5.2. Optimization Setup

We optimize separately for `gpt-5.4-mini` and `gpt-5.6-luna`, using `gpt-6-sol` to propose two candidates in each of five rounds. Candidates retain the same tools and limits of ten interaction turns, 4,000 characters, and twelve hits per observation. The project-level split assigns 828 requests to Train, 234 to Valid, and 289 to Test. Train supplies point-level diagnostics and Valid selects the incumbent; the selected policies are then frozen for fresh Test evaluations against GroupBrief. Two regenerated `gpt-5.4-mini` brief batches are reported separately on Test in the appendix. The seven Test projects are excluded from this search, although earlier experiments inspected Test, so it is not an untouched confirmatory set. Appendix E details the split and search controls.

5.3. Results and Analysis

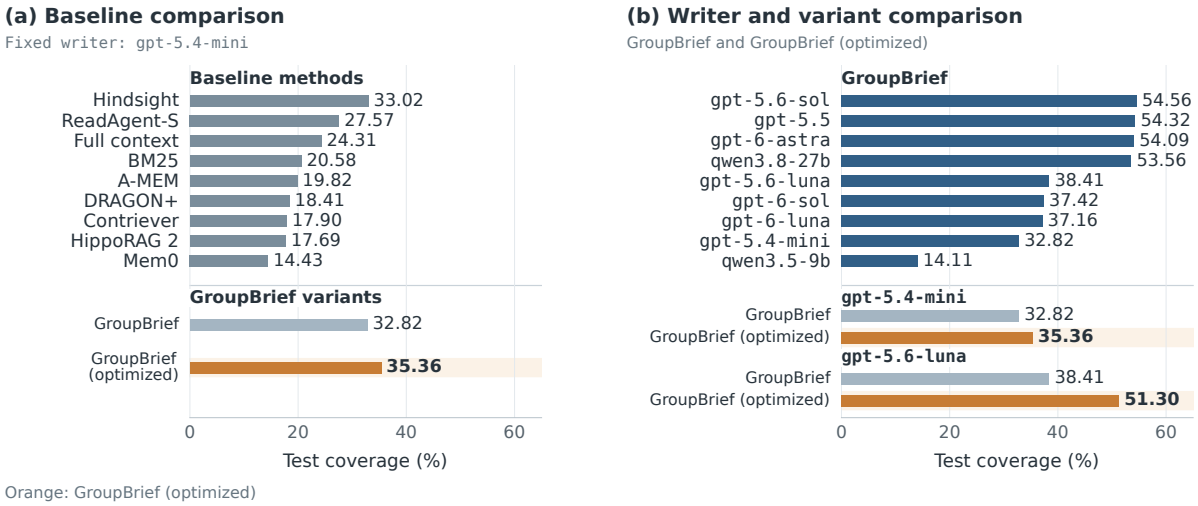


Figure 3 | Test coverage (%) on 289 requests and 2,162 reference points. Left: baseline methods and GroupBrief variants with gpt-5.4-mini. Right: nine writers using GroupBrief, followed by within-writer comparisons of GroupBrief and GroupBrief (optimized). Orange marks GroupBrief (optimized).

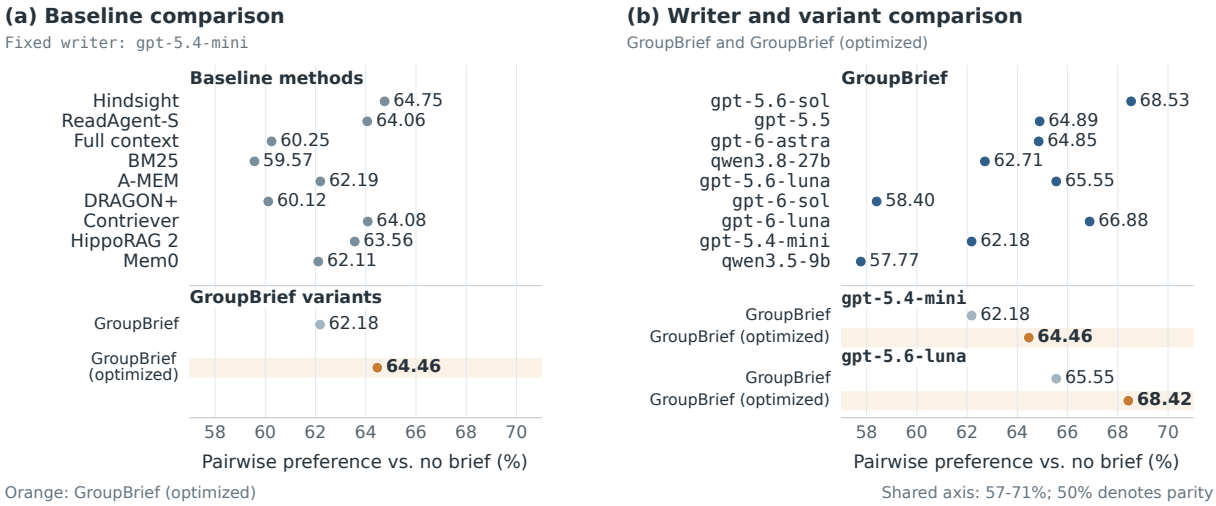


Figure 4 | Test simulation pairwise preference against no brief, with ties receiving half credit. Method order and colors match Figure 3. Both panels share a zoomed 57–71% axis; 50% denotes parity. Each rate uses its own valid comparisons and is not a direct judgment between GroupBrief and GroupBrief (optimized).

Policy optimization improves Test coverage with either writer. Figure 3 shows higher coverage after optimizing retrieval and synthesis. For gpt-5.4-mini, coverage increases from 32.82% to 35.36% (+2.54 pp); for gpt-5.6-luna, it increases from 38.41% to 51.30% (+12.88 pp). These fresh Test results use the policies selected in rounds 5 and 4, respectively. They show that the briefing procedure is an additional source of variation beyond writer choice, with a larger improvement for gpt-5.6-luna in this search.

Optimized briefs also provide useful collaboration context. In Test simulations, GroupBrief (optimized) obtains preference rates of 64.46% ($n = 242$) with gpt-5.4-mini and 68.42% ($n = 247$) with gpt-5.6-luna against no brief (Figure 4). The corresponding GroupBrief rates are 62.18% ($n = 238$) and 65.55% ($n = 254$). These results assess the selected briefs in interaction, beyond the coverage objective used during search. Each rate uses its own valid subset against the no-brief control; it is not a direct preference judgment between the two briefing variants. Appendix F.5 reports all Test configurations.

Most validation gains emerge early in search. Across the two searches, twenty proposals were evaluated and eighteen advanced to Valid. Figure 5 shows the retained Valid scores from GroupBrief through five rounds: 26.95, 32.37, 32.37, 34.84, 36.60, and 37.02% for `gpt-5.4-mini`, and 38.80, 44.70, 49.70, 49.79, 49.88, and 49.88% for `gpt-5.6-luna`. The latter changes little after the second round, whereas the former continues to improve in later rounds. These curves describe selection on the original search samples; their nondecreasing form follows the incumbent rule. The Test results above use fresh brief samples and exclude earlier Test draws. Appendix F.3.1 reports every candidate and its eligibility outcome.

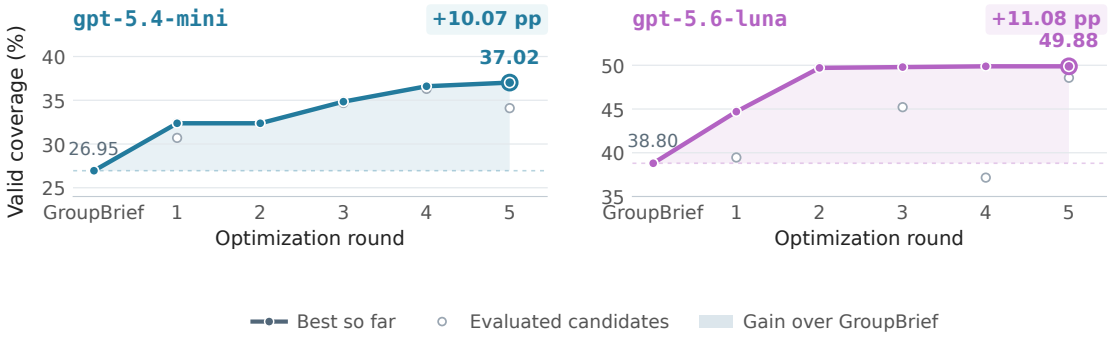


Figure 5 | Valid coverage during the five-round policy search, using the original search samples. The selected policies are subsequently evaluated on fresh brief samples. The nondecreasing incumbent curve follows the selection rule and shows larger gains in the early rounds.

Coordinator retrieval provides a separate diagnostic. To examine whether downstream evidence access changes the separation between briefing policies, we run an independent 40-task Train pilot with fresh trajectories and condition-matched no-brief controls. Removing coordinator search and reading preserves member communication and member-visible evidence. On the 22 tasks valid for both conditions and methods, the GroupBrief (optimized) minus GroupBrief preference gap is smaller without retrieval, with a difference of -6.82 pp. The no-brief control retrieves in only 3/40 tasks. This small pilot offers limited evidence about method separation and does not replace the main evaluation, which retains coordinator retrieval.

5.4. Harness Optimization versus Writer Choice

Writer choice and policy optimization offer two ways to change the resulting brief. Figures 3 and 4 compare them on the same 289 Test requests, with coverage scored against 2,162 reference points. Within each writer, the policy comparison holds the writer fixed; across writers, the comparison describes configurations with unmatched token budgets. The published memory baselines use `gpt-5.4-mini` and therefore supply no matched method comparison for `gpt-5.6-luna`. Coverage and simulation also measure different outcomes and may use different brief samples.

The primary comparisons use each writer's original GroupBrief brief batch and its corresponding GroupBrief (optimized) batch. Regenerated briefs illustrate why sample identity matters: for two additional `gpt-5.4-mini` batches, Test coverage increases from 34.14% to 36.96%, while simulation preference is 64.46% ($n = 242$) for GroupBrief and 62.25% ($n = 253$) for GroupBrief (optimized). These separate samples, reported in the appendix with their own valid sets, show that greater reference coverage need not coincide with higher observed downstream preference in every brief generation.

5.5. Coverage across the Collaboration Taxonomy

To identify which coordination information remains difficult, we break down Test coverage by four primary classes and their literature-based subtypes. Aggregation includes all 2,162 points,

including eleven in rare interpersonal, affective, and cooperation subtypes; Figure 6 displays the seven subtypes with at least ten points. We compare GroupBrief and GroupBrief (optimized) for `gpt-5.4-mini` and `gpt-5.6-luna`, alongside GroupBrief with `gpt-5.6-sol` and Hindsight with `gpt-5.4-mini`. These last two configurations have the highest observed full-benchmark coverage among GroupBrief writers and published fixed-writer baselines, respectively. Their inclusion is descriptive, with computation budgets unmatched.

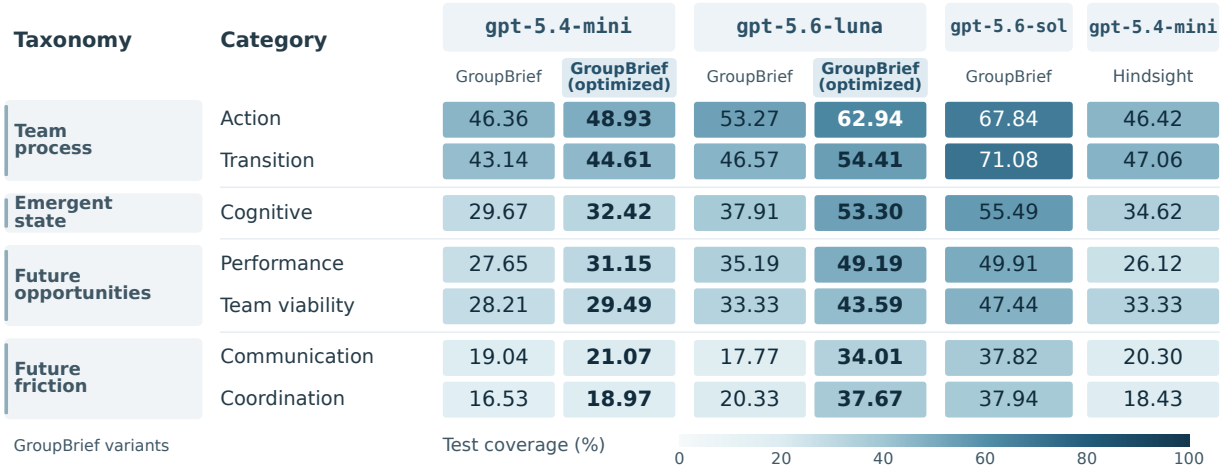


Figure 6 | Test coverage (%) for seven subtypes of the collaboration taxonomy on the same 289 tasks. Paired columns compare GroupBrief and GroupBrief (optimized) within each writer; the final two columns show `gpt-5.6-sol` with GroupBrief and Hindsight with `gpt-5.4-mini`. Subtypes with fewer than ten Test points are omitted from the display. Darker cells indicate higher coverage on a shared 0–100 scale.

Higher overall coverage leaves a persistent category gap. With GroupBrief fixed, `gpt-5.6-sol` reaches 54.56% Test coverage, compared with 32.82% for `gpt-5.4-mini`, and covers more information in all four primary classes. Its future-friction coverage nevertheless remains 37.94%, well below its 68.24% team-process coverage. Thus the gap between prospective problems and established coordination persists with a higher-coverage writer. The comparison is subject to same-model evaluation bias because `gpt-5.6-sol` also judges coverage.

Similar aggregate scores can conceal different information profiles. In the subtype-analysis sample, Hindsight with `gpt-5.4-mini` reaches 33.26% overall coverage, close to GroupBrief's 32.82%. It covers more emergent-state information (34.24% versus 30.43%) but fewer future opportunities (26.59% versus 27.68%). GroupBrief (optimized) reaches 35.36% overall with the same writer, yet remains below Hindsight on emergent state. Category-level analysis therefore reveals differences in the coordination context preserved by briefs that an overall score alone cannot express.

All category scores use point-weighted aggregation. Hindsight's subtype-analysis sample differs from its main aggregate coverage sample, and the rare subtypes omitted from the display contain too few points for robust ranking comparisons. These profiles describe the evaluated briefs without isolating the causal contribution of individual retrieval or synthesis components. We next examine an earlier case to make the policy instructions and resulting information differences concrete.

5.6. Case Study: What GroupBrief (optimized) Learns to Preserve

We use a healthcare case to examine what the revised policy asks the writer to preserve (Figure 7). GroupBrief and GroupBrief (optimized) share the same gpt-5.4-mini writer, tools, and budgets; only retrieval and writing instructions change. The optimized policy was selected on Valid in round 5 and frozen before Test. For context, we also examine GroupBrief with gpt-5.6-sol and Full context with gpt-5.4-mini on the same request and reference checklist.

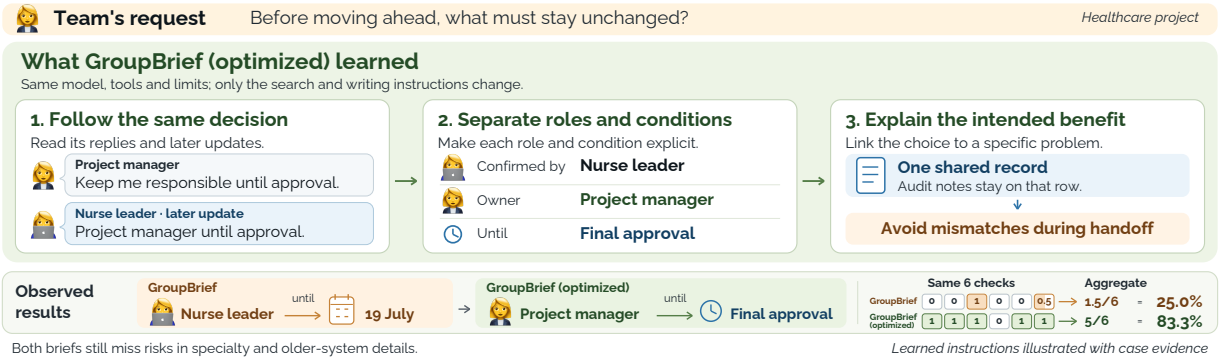


Figure 7 | Practices emphasized by GroupBrief (optimized) in a healthcare case. The main panels illustrate decision follow-up, separation of confirming and responsible roles, and preservation of conditions and purpose. The lower diagrams compare ownership statements and aggregate credit over the same six reference points. Messages are paraphrased and participants are labeled by role. Arrows illustrate policy instructions, not recorded tool calls; an approval condition does not establish completed approval.

Following decisions and preserving roles and conditions. GroupBrief (optimized) makes three existing capabilities of GroupBrief more explicit. It asks the agent to follow a decision through replies and later pre-cutoff mentions, preserving updates about the same item. It distinguishes the person confirming an arrangement from the person responsible for it, together with the condition limiting that responsibility. It also connects the arrangement to the problem it is intended to prevent. These instructions sharpen thread follow-up and the preservation of conditions and purpose; an intended benefit remains distinct from an achieved outcome.

Preserving the current owner, condition, and purpose. The original GroupBrief brief names the nurse leader as responsible through 19 July. GroupBrief (optimized) instead retains the latest confirmation: the project manager remains responsible until final approval, without implying that approval has occurred. It also explains why audit notes should stay in the same tracker row—to avoid a second record path and mismatches during handoff. Credit on the same six reference points rises from 1.5/6 (25.0%) to 5/6 (83.3%). Both briefs still omit risks involving specialty and legacy details outside the core map. The observed difference therefore concerns both the ownership relation and the coordination details surrounding it.

Ownership is only part of the required context. GroupBrief with gpt-5.6-sol scores 6/6 (100.0%) on the same task and checklist. Its brief contrasts earlier nurse-leader ownership with later project-manager ownership through sign-off, while retaining the source row, inline audit context, and traceable handoff. It also receives full credit for the specialty and legacy risks omitted by both gpt-5.4-mini briefs. Because gpt-5.6-sol serves as both writer and judge, this comparison is subject to same-model evaluation bias.

Full context with gpt-5.4-mini scores 4/6 (66.7%). It retains the current owner, single-source row, inline audit text, and single approval path, but misses the specialty and legacy risks and the complete traceable-handoff point. That point requires keeping the owner, seam, approval note, and validation flag together on the row, which GroupBrief (optimized) preserves. Full context

contains all 1,632 eligible messages without truncation and uses 168,403 final-writer input tokens, so truncation alone cannot explain these omissions. All three higher-coverage briefs recover current ownership; their remaining differences concern other coordination details.

This post-hoc Test case links revised instructions to observed brief differences. It neither reconstructs executed tool calls nor establishes a causal mechanism or downstream improvement. The additional comparisons use unmatched model-call counts and token budgets and support no general writer or method ranking, or conclusion that policy optimization is unnecessary.

6. Related Work

Long-term memory for language agents. Agent memory research studies how to retain experience and make it available for later use. Approaches include hierarchical storage and memory management (Kang et al., 2025; Packer et al., 2023; Zhong et al., 2024), structured extraction and associative retrieval (Chhikara et al., 2025; Jiménez Gutiérrez et al., 2024; Salama et al., 2025; Xu et al., 2025), and episodic organization and reflection (Latimer et al., 2025; Nan et al., 2025). Personalization methods also use histories to model users and adapt responses to individual preferences (Jiang et al., 2025; Salemi et al., 2024; Tan and Jiang, 2023; Tan et al., 2024a,b, 2026). Group briefing builds on these capabilities by asking how historical evidence should be selected and composed for a particular collaborative task. Its output connects participants' expressed knowledge, commitments, and dependencies in a brief for a collaborative planner. Persistent and personal memory can provide complementary information for this preparation.

Memory evaluation and multi-party state reconstruction. Memory benchmarks evaluate recall, temporal reasoning, knowledge revision, reflection, and incremental learning (Hu et al., 2026b; Maharana et al., 2024; Tan et al., 2025; Wu et al., 2025). Recent evaluations also test memory-guided actions and interdependent tasks (He et al., 2026; Shen et al., 2026). GroupMemBench (Yang et al., 2026), EverMemBench (Hu et al., 2026a), and SocialMemBench (Owolabi, 2026) extend this setting to speaker attribution, evolving knowledge, participant-specific context, and the application of remembered constraints. Related architectures preserve speaker-centered views, evidence provenance, and evolving conversational state (Lu et al., 2026; Zheng et al., 2026). These studies establish relational state and the use of memory beyond recall as important evaluation targets. We examine how to compile this state into preparation for a specific collaboration, preserving unresolved ownership, mismatched assumptions, and cross-participant dependencies before planning begins. Our benchmark evaluates the coverage of historical and prospective collaboration information, alongside its usefulness in downstream simulations.

Social reasoning and collaborative coordination. Preparing an agent for collaboration requires reasoning about both participants and their shared work. Theory-of-mind benchmarks test whether models distinguish participants' beliefs and information access (Kim et al., 2023; Wu et al., 2023). Social-agent research connects memory and reflection with interactive behavior (Park et al., 2023), while SOTOPIA (Zhou et al., 2024) and SOTOPIA-π (Wang et al., 2024) study its evaluation and learning. Human-agent collaboration research examines communication, shared workspaces, and coordination scaffolds (Kotalwar et al., 2026; Shao et al., 2026). Together, these directions connect to accounts of common ground, mutual knowledge, and organizational coordination (Clark and Brennan, 1991; Cramton, 2001; Okhuysen and Bechky, 2009). Our focus is the historical preparation available when an agent enters an ongoing collaboration: what participants have expressed, acknowledged, or committed to, and where agreement or ownership remains unresolved. Evaluating this preparation separately from interaction makes the agent's initial coordination context explicit.

7. Discussion and Limitations

The results support studying both the coordination information a brief preserves and how an agent uses it. These evaluations also have distinct limits. Coverage measures retention of reference information, without scoring the factual precision of additional claims or isolating value beyond personal memory. The corpus is synthetic, its labels are model-assisted, and simulation criteria draw on the same reference information as coverage. Model-generated member reports, compressed member evidence, and reuse of unchanged final reports further limit interaction fidelity. The two evaluations therefore provide complementary evidence within this benchmark; neither establishes effectiveness in unrestricted real collaboration.

Simulation preference is conditional on the execution and judging protocol. The bounded retry policy measures performance after recovery within a common attempt budget, rather than first-attempt robustness. Each marginal rate uses its own set of order-consistent outcomes; the paired analysis in the separate forty-task Train pilot uses common valid tasks. Two temperature-one presentations cannot distinguish systematic order effects from sampling variation. The pilot examines coordinator retrieval on a small diagnostic sample, while the main evaluation retains retrieval.

Optimization results should also be read in light of the selection process. Valid is reused across search rounds, Test was inspected in earlier experiments, and the Test split contains only seven projects. These conditions limit confirmatory claims. Finally, current writer and memory billing has not been reconciled, so historical cost ratios cannot establish the efficiency of the evaluated configurations. Broader validation should examine real teams, additional projects, factual precision, and fully reconciled resource use alongside coverage and interaction quality.

8. Conclusion

We introduced group briefing as preparation for agents entering ongoing collaboration: reconstructing task-relevant group cognition from the interaction history available at a specified cutoff. Our benchmark comprises 1,351 requests and 9,975 verified reference points and evaluates both information coverage and downstream interaction. The experiments reveal a persistent gap between recovering past state and identifying future friction, even for the highest-coverage writers. GroupBrief offers a task-guided retrieval and synthesis procedure for studying this challenge. With the writer fixed, coverage-guided policy optimization improves Test coverage by 2.54 percentage points for `gpt-5.4-mini` and 12.88 for `gpt-5.6-luna`; the optimized variants obtain simulation preferences of 64.46% and 68.42% against no brief on valid Test comparisons. These findings highlight the value of evaluating how interaction history is organized into collaboration context, including the responsibilities, conditions, and unresolved dependencies that an agent needs before joining the team's work.

References

Y. Bai, X. Lv, J. Zhang, H. Lyu, J. Tang, Z. Huang, Z. Du, X. Liu, A. Zeng, L. Hou, Y. Dong, J. Tang, and J. Li. LongBench: A bilingual, multitask benchmark for long context understanding. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 3119–3137. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.172. URL https://aclanthology.org/2024.acl-long.172/.

P. Chhikara, D. Khant, S. Aryan, T. Singh, and D. Yadav. Mem0: Building production-ready AI agents with scalable long-term memory, 2025. arXiv preprint arXiv:2504.19413.

H. H. Clark and S. E. Brennan. Grounding in communication. In L. B. Resnick, J. M. Levine, and S. D. Teasley, editors, *Perspectives on Socially Shared Cognition*, pages 127–149. American Psychological Association, 1991. doi: 10.1037/10096-006.

C. D. Cramton. The mutual knowledge problem and its consequences for dispersed collaboration. *Organization Science*, 12(3):346–371, 2001. doi: 10.1287/orsc.12.3.346.10098.

Z. He, Y. Wang, C. Zhi, Y. Hu, T.-P. Chen, L. Yin, Z. Chen, T. A. Wu, S. Ouyang, Z. Wang, J. Pei, J. McAuley, Y. Choi, and A. Pentland. MemoryArena: Benchmarking agent memory in interdependent multi-session agentic tasks, 2026. URL https://arxiv.org/abs/2602.16313.

C. Hu, T. Li, X. Gao, H. Chen, Y. Bai, D. Xu, T. Lin, X. Li, Y. Han, J. Pei, and Y. Deng. Evaluating long-horizon memory for multi-party collaborative dialogues. In *Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 9082–9090. Association for Computing Machinery, 2026a. doi: 10.1145/3770855.3817589.

Y. Hu, Y. Wang, and J. McAuley. Evaluating memory in LLM agents via incremental multi-turn interactions. In *The Fourteenth International Conference on Learning Representations*, 2026b. URL https://openreview.net/forum?id=DT7JyQC3MR.

G. Izacard, M. Caron, L. Hosseini, S. Riedel, P. Bojanowski, A. Joulin, and E. Grave. Unsupervised dense information retrieval with contrastive learning. *Transactions on Machine Learning Research*, 2022. URL https://openreview.net/forum?id=jKN1pXi7b0.

B. Jiang, Y. Yuan, M. Shen, Z. Hao, Z. Xu, Z. Chen, Z. Liu, A. R. Vijjini, J. He, H. Yu, R. Poovendran, G. Wornell, L. Ungar, D. Roth, S. Chen, and C. J. Taylor. Personamem-v2: Towards personalized intelligence via learning implicit user personas and agentic memory, 2025. URL https://arxiv.org/abs/2512.06688.

B. Jiménez Gutiérrez, Y. Shu, Y. Gu, M. Yasunaga, and Y. Su. HippoRAG: Neurobiologically inspired long-term memory for large language models. In *Advances in Neural Information Processing Systems*, volume 37, 2024. doi: 10.52202/079017-1902.

B. Jiménez Gutiérrez, Y. Shu, W. Qi, S. Zhou, and Y. Su. From RAG to memory: Non-parametric continual learning for large language models. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pages 21497–21515. PMLR, 2025. URL https://proceedings.mlr.press/v267/gutierrez25a.html.

J. Kang, M. Ji, Z. Zhao, and T. Bai. Memory OS of AI agent. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 25961–25970. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.1318.

H. Kim, M. Sclar, X. Zhou, R. Le Bras, G. Kim, Y. Choi, and M. Sap. FANToM: A benchmark for stress-testing machine theory of mind in interactions. In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 14397–14413. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.emnlp-main.890.

N. Kotalwar, R. Das, and C. Rose. Searching for synergy in shared workspace human-AI collaboration, 2026. URL https://arxiv.org/abs/2606.18413.

C. Latimer, N. Boschi, A. Neeser, C. Bartholomew, G. Srivastava, X. Wang, and N. Ramakrishnan. Hindsight is 20/20: Building agent memory that retains, recalls, and reflects, 2025. URL https://arxiv.org/abs/2512.12818.

K.-H. Lee, X. Chen, H. Furuta, J. Canny, and I. Fischer. A human-inspired reading agent with gist memory of very long contexts. In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 26396–26415. PMLR, 2024. URL https://proceedings.mlr.press/v235/lee24c.html.

Z. Li, C. Li, M. Zhang, Q. Mei, and M. Bendersky. Retrieval augmented generation or long-context LLMs? a comprehensive study and hybrid approach. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, pages 881–893.

Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.emnlp-industry.66. URL https://aclanthology.org/2024.emnlp-industry.66/.

S.-C. Lin, A. Asai, M. Li, B. Oguz, J. Lin, Y. Mehdad, W.-t. Yih, and X. Chen. How to train your DRAGON: Diverse augmentation towards generalizable dense retrieval. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 6385–6400. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.findings-emnlp.423. URL https://aclanthology.org/2023.findings-emnlp.423/.

Z. Lu, Z. Yuan, L. Ou, and L. Shi. Propose, verify, commit: Evidence-grounded memory for long-horizon multi-actor conversations, 2026. URL https://arxiv.org/abs/2609.23465.

A. Maharana, D.-H. Lee, S. Tulyakov, M. Bansal, F. Barbieri, and Y. Fang. Evaluating very long-term conversational memory of LLM agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 13851–13870. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.747.

J. Nan, W. Ma, W. Wu, and Y. Chen. Nemori: Self-organizing agent memory inspired by cognitive science, 2025. URL https://arxiv.org/abs/2508.03341.

G. A. Okhuysen and B. A. Bechky. Coordination in organizations: An integrative perspective. *Academy of Management Annals*, 3(1):463–502, 2009. doi: 10.5465/19416520903047533.

O. Owolabi. SocialMemBench: Are AI memory systems ready for social group settings?, 2026. arXiv preprint arXiv:2605.17789.

C. Packer, S. Wooders, K. Lin, V. Fang, S. G. Patil, I. Stoica, and J. E. Gonzalez. MemGPT: Towards LLMs as operating systems, 2023. URL https://arxiv.org/abs/2310.08560.

J. S. Park, J. C. O'Brien, C. J. Cai, M. R. Morris, P. Liang, and M. S. Bernstein. Generative agents: Interactive simulacra of human behavior. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. Association for Computing Machinery, 2023. doi: 10.1145/3586183.3606763. URL https://doi.org/10.1145/3586183.3606763.

S. Robertson and H. Zaragoza. The probabilistic relevance framework: BM25 and beyond. *Foundations and Trends in Information Retrieval*, 3(4):333–389, 2009. doi: 10.1561/1500000019. URL https://doi.org/10.1561/1500000019.

R. Salama, J. Cai, M. Yuan, A. Currey, M. Sunkara, Y. Zhang, and Y. Benajiba. MemInsight: Autonomous memory augmentation for LLM agents. In *Proceedings of the 2025 Conference on Empirical Methods in Natural Language Processing*, pages 33136–33152. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.emnlp-main.1683.

A. Salemi, S. Mysore, M. Bendersky, and H. Zamani. LaMP: When large language models meet personalization. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 7370–7392. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.399.

Y. Shao, V. Samuel, Y. Jiang, J. Yang, and D. Yang. Collaborative gym: A framework for enabling and evaluating human-agent collaboration. In *The Fourteenth International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2412.15701.

Y. Shen, K. Li, W. Zhou, and S. Hu. Mem2ActBench: A benchmark for evaluating long-term memory utilization in task-oriented autonomous agents. In *Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 8173–8190. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.acl-long.370.

H. Tan, Z. Zhang, C. Ma, X. Chen, Q. Dai, and Z. Dong. MemBench: Towards more comprehensive evaluation on the memory of LLM-based agents. In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 19336–19352. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.findings-acl.989.

Z. Tan and M. Jiang. User modeling in the era of large language models: Current research and future directions. *IEEE Data Engineering Bulletin*, 46(4):57–96, 2023. URL https://arxiv.org/abs/2312.11518.

Z. Tan, Z. Liu, and M. Jiang. Personalized pieces: Efficient personalized large language models through collaborative efforts. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6459–6475. Association for Computational Linguistics, 2024a. doi: 10.18653/v1/2024.emnlp-main.371.

Z. Tan, Q. Zeng, Y. Tian, Z. Liu, B. Yin, and M. Jiang. Democratizing large language models via personalized parameter-efficient fine-tuning. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing*, pages 6476–6491. Association for Computational Linguistics, 2024b. doi: 10.18653/v1/2024.emnlp-main.372.

Z. Tan, X. Zhai, Y. Zhu, M. Jiang, and M. Hammad. Bridging textual profiles and latent user embeddings for personalization, 2026. arXiv preprint arXiv:2605.06981.

R. Wang, H. Yu, W. Zhang, Z. Qi, M. Sap, Y. Bisk, G. Neubig, and H. Zhu. SOTOPIA-π: Interactive learning of socially intelligent language agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 12912–12940. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.698.

D. Wu, H. Wang, W. Yu, Y. Zhang, K.-W. Chang, and D. Yu. LongMemEval: Benchmarking chat assistants on long-term interactive memory. In *The Thirteenth International Conference on Learning Representations*, 2025. URL https://proceedings.iclr.cc/paper_files/paper/2025/file/d813d324dbf0598bbdc9c8e79740ed01-Paper-Conference.pdf.

Y. Wu, Y. He, Y. Jia, R. Mihalcea, Y. Chen, and N. Deng. Hi-ToM: A benchmark for evaluating higher-order theory of mind reasoning in large language models. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, pages 10691–10706. Association for Computational Linguistics, 2023. doi: 10.18653/v1/2023.findings-emnlp.717.

P. Xu, W. Ping, X. Wu, L. McAfee, C. Zhu, Z. Liu, S. Subramanian, E. Bakhturina, M. Shoeybi, and B. Catanzaro. Retrieval meets long context large language models. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=xw5nxFWMlo.

W. Xu, Z. Liang, K. Mei, H. Gao, J. Tan, and Y. Zhang. A-Mem: Agentic memory for LLM agents. In *Advances in Neural Information Processing Systems*, volume 38, 2025. doi: 10.52202/085713-0593. URL https://proceedings.neurips.cc/paper_files/paper/2025/hash/19909c36f51abc4856b4560aff3d36d6-Abstract-Conference.html.

J. Yang, K.-H. Lai, X. Wang, S. Chang, Y. Harari, and E. Gabrilovich. GroupMemBench: Benchmarking LLM agent memory in multi-party conversations, 2026. URL https://arxiv.org/abs/2605.14498. arXiv preprint arXiv:2605.14498.

H. Zheng, T. Tang, Y. Chen, W. Wang, and Y. Wu. SpeakerMem-R1: Speaker-centered dual-track memory for multi-party dialogue, 2026. arXiv preprint arXiv:2609.26780.

M. Zhong, D. Yin, T. Yu, A. Zaidi, M. Mutuma, R. Jha, A. H. Awadallah, A. Celikyilmaz, Y. Liu, X. Qiu, and D. Radev. QMSum: A new benchmark for query-based multi-domain meeting summarization. In *Proceedings of the 2021 Conference of the North American Chapter of the*

Association for Computational Linguistics: Human Language Technologies, pages 5905–5921. Association for Computational Linguistics, 2021. doi: 10.18653/v1/2021.naacl-main.472. URL https://aclanthology.org/2021.naacl-main.472/.

W. Zhong, L. Guo, Q. Gao, H. Ye, and Y. Wang. MemoryBank: Enhancing large language models with long-term memory. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19724–19731, 2024. doi: 10.1609/aaai.v38i17.29946.

X. Zhou, H. Zhu, L. Mathur, R. Zhang, Z. Qi, H. Yu, L.-P. Morency, Y. Bisk, D. Fried, G. Neubig, and M. Sap. SOTOPIA: Interactive evaluation for social intelligence in language agents. In *The Twelfth International Conference on Learning Representations*, 2024. URL https://openreview.net/forum?id=mM7VurbA4r.

Appendix

A. Dataset Construction

Interaction histories and requests. To represent collaboration across different work settings, we generate synthetic group interactions in manufacturing, finance, healthcare, and technology. Histories are organized by project, topic, phase, and participant role. A knowledge graph stores personas, project structures, participant assignments, and phase-level timestamps. A Director plans exchanges, a Scene Writer produces them, and speaker-specific rewriting differentiates participants' voices. Validated messages update the graph and phase memory. Dialogue acts, intents, phase states, and decision-change metadata support later annotation checks. The resulting corpus represents generated interactions rather than observed workplace behavior.

Requests are constructed from the source state-query benchmark's existing project, phase, cutoff, and eligible-history snapshots. For each snapshot, `gpt-5.5` reads up to the 400 most recent eligible messages and attempts to generate two requests about distinct work items. Each request uses 7–20 English words to describe an upcoming activity, such as preparing a handoff or reviewing a decision, while withholding the facts needed to address it. Generation excludes internal message identifiers, numbers, structured enumerations, and generic conversation-summary requests. The target is two requests per snapshot; verification determines how many are released.

Initial reference construction and verification. The request-generation call also proposes 4–10 atomic reference points per request. Each records an assertion, work object, actors and roles, category, cutoff status, 1–4 core facets, 0–3 supporting facets, and evidence identifiers. Verification proceeds in two stages. Deterministic checks validate formatting, category–status compatibility, identifiers, and duplication. A separate model call then checks that each point is supported, current, necessary for the task, absent from the request, atomic, and correctly categorized; the request must also be realistic, answerable, and retain enough valid points. Invalid generations may be retried using deterministic feedback, and rejected points and verification decisions are retained. These separate calls are neither independent model-family verification nor human review, and generating requests and points jointly may introduce shared selection bias.

Reference points cover team process, team emergent state, future friction, and future opportunities. The first two describe arrangements, handoffs, and cognitive, affective, or motivational team states; we combine them as *past state* when reporting three groups. Released past-state points come from this initial construction, and future points come from the separately verified procedure below.

Prospective points and temporal validity. Prospective reference points connect a signal available at the cutoff to a later corroborating event. For future friction, the signal concerns an obstacle that subsequently materializes; for a future opportunity, it concerns an intended benefit that is later realized. Acceptance requires historical support, subsequent realization for the same object, relevance to the request, and no hindsight in the scored assertion or facets. Later adoption, operationalization, measurement, independent confirmation, or resolution can validate a point. The scored text nevertheless remains limited to what was knowable at the cutoff.

The first mining round examines up to 400 eligible pre-cutoff messages and the 400 earliest non-noise post-cutoff messages. A second round expands both windows to 800 messages, excludes previously discovered points, and checks novelty. Separate verification calls assess historical support, realization, absence of hindsight, task necessity, atomicity, and category fit. A round may yield no additional points, and no category is filled solely to meet a quota.

We keep pre-cutoff evidence, later evidence, and the typed validation relation in separate fields. Later evidence supports construction only and is withheld from the writer and coverage judge. Assertions and facets describe concerns or intended benefits identifiable at the cutoff. Consequently,

Table 3 | Task-oriented briefing dataset. Past state combines team process and team emergent state. Future categories contain pre-cutoff signals corroborated by subsequent interaction.

Domain	Queries	Points	Past state	Friction	Opportunities
Manufacturing	361	2,606	1,230	672	704
Finance	286	2,133	964	559	610
Healthcare	382	2,789	1,277	731	781
Technology	322	2,447	1,105	627	715
Total	1,351	9,975	4,576	2,589	2,810

the labels capture subsequently corroborated collaboration information and favor signals realized within the available follow-up window; they are not labels for open-world future-event prediction.

Metadata verification and release selection. Metadata verification checks that a candidate remains consistent with the generated interaction state. The evidence package combines cited messages with dialogue acts, intents, phase states, decision labels, change records, and messages linked through replies, references, or decision changes. Deterministic warnings flag possible temporal problems. A separate `gpt-5.5` call requires four checks to pass: metadata consistency, temporal status, absence of supersession, and confirmation of the future event where applicable.

Of 16,529 candidates checked at this stage, 13,487 pass. Release-time density control retains at most five points from each future group, rotating across subtypes. For retained friction and opportunity counts n_f, n_o, the past-state cap is $\max(3, \max(n_f, n_o) + 1)$, with no minimum of three past-state points. Future candidates are deduplicated by normalized assertion within each request. Each released request must retain at least five verified points spanning at least two reporting groups. No points are invented to meet these conditions. The resulting release contains 1,351 requests and 9,975 points, averaging 7.38 points per request (Table 3). Candidate, metadata-acceptance, and release counts refer to different construction stages. Because multiple requests may cite the same messages, reference points are task-specific propositions and need not be independent facts.

Frozen records and information boundaries. Frozen records make the task and its information boundary reproducible. The release links task records, request-specific gold points, and project histories. Each task inherits its project, phase, cutoff, and visibility selectors from a source snapshot and receives a new request and task identifier; gold-point identifiers refer only to that request's labels. Manifests record construction versions, density settings, and artifact hashes. All writers and baselines use the same frozen tasks, cutoffs, and points.

At generation time, the writer receives the public project/topic, phase, cutoff, and request, plus eligible messages containing only their identifier, author, timestamp, and content. Gold identifiers, semantic facets, verification decisions, generator metadata, and post-cutoff messages are excluded. The 400/800-message windows apply to annotation. Each method can access the complete eligible history within its own retrieval and context budgets. Memory banks and indexes are isolated by project.

For coverage, the judge receives only seven semantic fields from a point: taxonomy path, assertion, object, actors or roles, status, core facets, and supporting facets. It sees neither construction evidence nor the source conversation. In simulation, hidden gold-derived constraints and pre-cutoff evidence are provided separately to the pairwise judge, not to the coordinator or members. The coordinator receives the request, public follow-up, permitted context, and optional private brief. Members initially receive only the historical messages they authored and subsequently receive information delivered to them. They have no automatic access to the archive, brief, peers' private reports, or hidden labels.

B. Implementation Details

The coverage experiments with GroupBrief use nine writers: `gpt-5.6-sol`, `qwen3.8-27b`, `gpt-5.5`, `gpt-6-astra`, `gpt-5.6-luna`, `gpt-6-luna`, `gpt-6-sol`, `gpt-5.4-mini`, and `qwen3.5-9b`. All use the same 1,351 requests and 9,975 points. Model names follow the deployment or served-model identifiers in the experimental records. The historical `v6` retrieval-and-writing policy is a separate comparison and was neither an iteration nor a selected policy in the current harness search.

Policies implement `ReadPolicy` on a frozen `AgenticSubstrate`, which provides ingestion, indexing, tool dispatch, retrieval, and citation cleaning. Available tools are `list_participants`, BM25 `search`, `grep`, `open_thread`, `reply_chain`, `by_author`, `by_mention`, `time_window`, and `open_message`. GroupBrief requires at least two tool calls before accepting `finalize`, followed by a separate synthesis call; sections without supporting content state that no relevant information was identified. Optimization holds the budgets fixed at ten interaction turns, 4,000 characters per tool observation, and twelve messages per observation. Brief length, actual token consumption, and latency may still differ.

C. Evaluation Details

Judge and execution models. The evaluation in Section 2.4 uses `gpt-5.6-sol` for both coverage judging and pairwise simulation judging. The downstream execution model is fixed to `gpt-5.4-mini`. Using `gpt-5.6-sol` as both a writer and coverage judge may introduce same-model evaluation bias.

Facet checks and deterministic credit. Coverage judging separates semantic checks from numerical credit. For each task and method, the same saved brief is evaluated against every associated point. The judge receives the point's seven-field semantic view and the original nonempty brief lines, each with an identifier. It labels the object, actor/role scope, temporal status, and every core and supporting facet as `supported`, `missing`, or `contradicted`. Supported and contradicted judgments must cite original line identifiers; missing judgments must cite none. The decoder validates the identifiers, extracts the original text, and checks that each facet occurs exactly once. Paraphrases can provide support, but topical overlap or partial coverage of a compound facet cannot.

The judge does not choose point credit directly. The deterministic rule in Section 2.4.1 assigns zero for missing object or status, scope contradiction, any unsupported core facet, or any contradicted supporting facet. Otherwise, a missing supporting facet yields 0.5 and complete support yields 1. Missing actor scope alone does not force zero, but a core facet requiring a named actor must be supported. Points without supporting facets receive full credit when their scope and core checks pass.

Overall, domain, and category scores weight gold points equally within the relevant set, rather than averaging task percentages. Malformed, incomplete, refused, or unevidenced judge responses are retried under the recorded protocol. Errors that persist after retries are flagged and receive zero credit, with the gold-point denominator unchanged. Writer failures are recorded as failed tasks rather than successful empty briefs. Coverage measures whether the brief communicates the semantic checklist; it does not separately verify additional claims against the source conversation or measure factual precision, hallucination, brevity, or downstream utility.

Scenario construction and matched controls. A simulation case is attached to a released task and reuses its source snapshot's project, cutoff, eligible history, active members and roles, and recent-message identifiers. The coordinator receives the task request and the fixed public follow-up, "Can you check with whoever needs to weigh in and help me get this right?" For each gold point, the evaluator separately records its object, pre-cutoff evidence, cutoff constraint, affected decision,

and an observable coordination-progress criterion. Later events used in gold construction are not injected into the simulated conversation. Public scenario fields are restricted to project, cutoff, original request, follow-up, members, history, and recent-message identifiers. Scenario preparation verifies the task–scenario match and the eligibility of every public message.

All 1,351 requests have verified scenarios. Evaluation uses replicate 0. The baseline and writer summaries in Tables 1 and 2 cover the full benchmark; final comparisons involving GroupBrief (optimized) use the 289 Test tasks. The API seed is derived from the case and replicate identifiers, so replicate 0 does not assign literal API seed zero to every request. Unavailable-scenario counts from historical runs are excluded.

Interaction and member information. The matched simulation holds the starting information and action budget fixed. Both conditions begin with the same private initial member reports, which saved-brief experiments reuse along with the completed no-brief trajectory. Each brief-conditioned trajectory receives the exact original brief and allows at most four action steps, including three contact actions. Search and reading draw from this same budget. A contact can address several recipients, so three contacts need not mean three people or three replies.

Members respond using their own visible evidence and received messages. A source identifier mentioned in the private brief does not imply delivery of that source text. Each member report contains an interpretation of at most 80 words, an intended action of at most 60 words, an optional time window of at most 20 words, and at most four permitted evidence identifiers. The simulator selects an approximately 4,096-token actor-visible evidence packet, retaining required evidence on overflow. At final reporting, a validated report may be reused only if the member's received information is unchanged; otherwise, a new private report is generated. Reports are withheld from the coordinator and peers and describe understanding and intended next steps rather than externally completed work.

Evidence-grounded, counterbalanced judging. Pairwise judging assesses whether the interaction supports evidence-grounded coordination. The judge receives anonymized public trajectories and clearly marked private initial and final reports, while method labels, the brief, and coverage scores are withheld. It considers evidence fidelity, authority boundaries, participant uptake, compatibility of intended actions, and unnecessary interaction. Under these criteria, a justified unresolved blocker can be preferable to unsupported agreement.

The judge's pre-cutoff evidence packet has a 160-message budget. If the history exceeds this budget, it retains the latest 20 messages and explicit references, then fills remaining slots using BM25 over the request and trajectory text, adding adjacent messages when space permits. Explicit references exceeding the budget cause an error rather than silent deletion. Absence from this evidence subset is not proof that a claim is false.

The judge directly compares the two trajectories in AB and BA display orders while preserving stable anonymous identities. No audit, verification-model, or tiebreaker call is added. A stable win, loss, or tie requires agreement between both valid judgments. Disagreements are classified as order-sensitive, abstentions as unjudgeable, and failed presentations as judge failures. Original raw responses are retained.

Controls, exclusions, and uncertainty. Before simulation, bounded member-update controls and nine blind-judge calibration fixtures must pass. These checks test protocol behavior without establishing the realism of open-ended member reports. Unavailable scenarios, failed initialization or interaction, failed brief preparation, judge/API or validation failures, order-sensitive outcomes, and unjudgeable outcomes are retained as exclusions where produced. They are not converted into wins, ties, or losses. Pending cases remain distinct from processed exclusions. The denominator in

Eq. 4 contains only stable, valid win/loss/tie outcomes, so rates are conditional preferences, not unconditional task success; differing failure rates may bias comparisons.

Test preference rates describe each method's own valid query/seed set. To estimate a paired difference, methods must instead be compared on the intersection of those sets, subtracting their win/tie/loss scores for each common case against the same no-brief trajectory. Differences between the reported marginal rates therefore are not paired effects. The independent coordinator-retrieval pilot reports its within-pilot paired contrast separately.

Saved artifacts and offline verification. Run plans record the dataset, source hashes, method code, model identifiers, decoding settings, and planned inventory. Saved predictions, raw judgments, and facet verdicts are linked to the task and brief digest. The November 1 archive also retains complete attempt histories and policy-selection provenance. Offline verification re-decodes original-line evidence, recomputes credit, and checks exact task–gold membership without further judge calls. Simulation verification checks matched inputs, action budgets, controls, and the reduction of the two presentation-order judgments. Completion records enumerate processed outcomes, including failures.

Compatible cached responses may be reused. Retries, repeated simulation seeds, alternative judge presentations, and offline replay are recorded as distinct operations. Simulation uses saved briefs unchanged, including harness briefs recovered from generation traces. Recovery preserves completed predictions and outcomes while retaining explicit records of infrastructure failures. All rebuilt coverage points were scored without judge errors. Failed attempts remain archived, and reported results follow the uniform bounded retry policy. We report coverage and downstream results separately because the coordinator may misuse retained information or recover omitted information through further interaction.

D. Baseline Implementations and Resource Accounting

Common interface and comparison scope. The baseline comparison uses `gpt-5.4-mini` as the writer and `gpt-5.6-sol` as the coverage judge for the same 1,351 queries and 9,975 gold points. Each method receives only the query's eligible history, denoted here by $V_T = \mathcal{H}_g^{\leq T}$, and supplies evidence or memory to a final brief-writing call. Final synthesis uses GroupBrief's synthesis instruction, per-query sampling seed, and temperature 1; citations outside V_T are removed using the same cleaning procedure. Baseline synthesis also repeats the request, whereas GroupBrief's final synthesis prompt does not. This prompt difference and the unequal computation budgets limit attribution of performance differences to evidence selection alone. Ingestion and other model-dependent baseline operations use `gpt-5.4-mini`.

Full context (Li et al., 2024). We follow the long-context setting, rendering all eligible messages with GroupBrief's message headers for a single synthesis call. When the input exceeds the configured writer limit, we remove the middle of the history, following the truncation convention of LongBench (Bai et al., 2024).

BM25 (Robertson and Zaragoza, 2009), Contriever (Izacard et al., 2022), and DRAGON+ (Lin et al., 2023). Following Li et al. (2024); Xu et al. (2024), we divide histories into 300-word chunks and present the five highest-ranked chunks in score order with their indices. Dense retrieval uses cosine similarity with either `facebook/contriever-msmarco` or the DRAGON+ query and context encoders. The BM25 variant uses the same chunks and top-five budget, the tokenizer from GroupBrief's search tool, and $k_1 = 1.5$, $b = 0.75$.

ReadAgent-S (Lee et al., 2024). We use the authors' released QMSum prompts (Zhong et al., 2021), pagination limits of 600/280 words, gist memory, and sequential lookup of at most six pages. Pagination, gisting, and lookup use the released GPT settings, temperature 0 and a 512-token output limit; final brief synthesis uses the common settings above. The source notebook is fixed at commit `569dff3e`. A regular-expression parser additionally accepts page selections formatted as `Label: <N>`, which the notebook's original parser misses.

A-MEM (Xu et al., 2025). We use the authors' `memory_layer.py` and `test_advanced.py` from AgenticMemory commit `0c8039f2`, with one note per message in LoCoMo turn format (Maharana et al., 2024). We set `retrieve_k=40`, one of the two `gpt-4o-mini` values in the paper's hyperparameter table, rather than the repository default of 10. The controller passes the otherwise identical request using `max_completion_tokens`, as required by the deployment, and skips model-proposed link indices outside the memory list instead of raising an indexing error. If retrieved notes and linked neighbors exceed the writer limit, whole trailing entries are dropped.

Hindsight (Latimer et al., 2025). We use the official Hindsight 1.0.0 server (commit `5331da2f`), with its default local embedding model and cross-encoder reranker. Fact extraction, entity linking, and observation consolidation use `gpt-5.4-mini`. The authors' Agent Memory Benchmark harness, commit `aef3702e`, supplies the retention and recall protocol for LoCoMo and LongMemEval (Maharana et al., 2024; Wu et al., 2025): a session is retained as one timestamped document of JSON turns, instantiated here as one project-day of chat. All retention and consolidation operations finish before a cutoff is queried. Recall uses budget `high`, 32,768 memory tokens, source chunks up to 16,384 tokens, and the cutoff as query time. The harness's own deduplication and document-formatting functions render memories and source excerpts. PostgreSQL 18 runs as an unprivileged user, with eight API processes and separate workers sharing one database. After the initial restart into this distributed deployment, retention resumed for interrupted cutoffs using the same document identifiers, replacing existing records without duplication.

HippoRAG 2 (Jiménez Gutiérrez et al., 2025) and Mem0 (Chhikara et al., 2025). We adapt the multi-party chat protocol of GroupMemBench (Yang et al., 2026): each message is ingested in chronological order as a speaker-prefixed memory unit, embeddings use `text-embedding-3-large`, the query combines the asking user and request, and the top ten units are shown under source-message headers containing speaker, channel, timestamp, reply target, and message identifier. Ingestion uses `gpt-5.4-mini` in place of GroupMemBench's `gpt-4o-mini`. GroupMemBench commit `e2682e01` releases passage formatting and query construction, but does not release its HippoRAG and Mem0 runners.

HippoRAG 2 uses official `hipporag` 2.0.0a4: named-entity and triple extraction on new passages, a graph rebuilt at each cutoff, recognition-memory filtering, and personalized PageRank. We make two compatibility and error-handling changes. Its entity parser also accepts the bare JSON list requested by its prompt, in addition to the example's `named_entities` object. Its client honors the configured 40-attempt retry budget instead of using two attempts, and an extraction still failing after retries raises an error instead of silently indexing an empty result. These compatibility changes leave the computational budgets unequal across methods.

Mem0 uses official `mem0ai` 2.2.1, with spaCy entity linking and BM25 components from its benchmark server (memory-benchmarks commit `4b61c5d3`), additive extraction, and one `add` call per message as in its LoCoMo benchmark. Rate-limit exceptions are re-raised for caller retries, following its benchmark client.

Temporal isolation and resource scope. All systems isolate data by project and enforce each task's cutoff. Current coverage results use the completed rebuild, which is a different sample

Table 4 | Project-level data split and the role of each partition in harness optimization.

Split	Queries	Points	Role
Train	828	6,134	Candidate evaluation and diagnostics
Valid	234	1,679	Repeated policy selection
Test	289	2,162	One evaluation after policy freezing

from the earlier memory-cost audit. GroupBrief, Full context, and memory methods have different context and construction budgets. Historical per-query token ratios, truncation counts, and memory-operation totals therefore do not describe the current sample. Appendix F.2 specifies the resource-accounting scope, and Appendix F.5 reports the Test simulation denominators.

E. Search Configuration and Data Splits

The search in Section 5.1 optimizes coverage separately for `gpt-5.4-mini` and `gpt-5.6-luna`. Both searches use Codex with `gpt-6-sol` at medium reasoning effort as the proposer and complete five rounds with two candidates per round. Candidates must pass Python compilation, inheritance-interface checks, frozen-component checks, and budget validation before evaluation. A Landlock sandbox restricts proposer filesystem access, and unchanged candidates reuse cached evaluations. Runtime concurrency is two, increasing to four for the last `gpt-5.6-luna` rounds, without affecting selection or scoring. Coverage is the sole evaluation objective and selection signal; inference-time outcomes, simulation losses, and disagreements do not inform the search.

We partition instances by project using `domain_project_holdout_v1` with split seed 0 (Table 4). The search uses all 828 training requests. Train provides candidate scores and point-level diagnostics, while Valid provides aggregate feedback for repeated incumbent selection. Date ranges may overlap across splits, although each request enforces its own cutoff. Project separation reduces reuse of same-project context but does not eliminate shared generation templates or cross-project relationships.

GroupBrief (optimized) uses the policy selected in round 5 for `gpt-5.4-mini` and round 4 for `gpt-5.6-luna`. Both policies were frozen before the historical Test evaluation of GroupBrief and GroupBrief (optimized), conducted once per writer. Test contains seven projects, 289 requests, and 2,162 points. Repeated validation use introduces selection bias. Test was excluded from this search but inspected in historical experiments, so it is not an untouched confirmatory set. The current downstream evaluations for `gpt-5.4-mini` and `gpt-5.6-luna` cover all Test tasks. Figures 3 and 4 and Appendix F.5 report these current evaluations, separately from the historical search samples.

The eligibility tolerance is $\epsilon = 0.02$ on the unit-interval coverage scale, i.e., two percentage points below GroupBrief on each split. Appendix F.3.1 reports all candidate evaluations and their gate outcomes.

F. Additional Results

F.1. Simulation Reporting and Retry Policy

Final performance comparisons involving GroupBrief (optimized) use the 289 Test tasks from seven projects. Table 9 reports each method's valid denominator and conditional preference against no brief. The full-benchmark baseline and writer summaries in Tables 1 and 2 are descriptive comparisons separate from this Test evaluation.

Uniform bounded retry policy. A failed trajectory receives at most two additional whole-trajectory attempts, and the first complete trajectory is retained regardless of its subsequent

judgment. When a failed shared no-brief trajectory is recovered, compatible completed treatment trajectories are reused and rejudged against that recovered control. Earlier attempt records remain archived unchanged. Exact-request transport recovery may repeat a failed judge presentation while preserving the successful opposite-order verdict. Completed trajectories are not regenerated, and disagreements and abstentions are not retried to obtain agreement. Reported preferences describe results under this attempt budget rather than first-attempt robustness.

F.2. Current Coverage and Resource Scope

Current coverage results use the rebuilt briefs, with reporting scope determined by the comparison. Tables 1 and 2 summarize 1,351 briefs and 9,975 point judgments per configuration without scoring errors. Final GroupBrief–GroupBrief (optimized) comparisons use the 289 Test requests and 2,162 reference points in Figure 3; Table 8 separately reports regenerated `gpt-5.4-mini` samples. Coverage is point-weighted, and historical token ratios do not transfer to these new samples.

Resource-accounting scope. Resource accounting must distinguish brief construction from downstream evaluation. The simulation results include bounded trajectory retries and exact-request transport recovery, but no complete billing audit or globally deduplicated call count across these stages. Brief generation, offline memory construction, shared initial-state preparation, cached requests, and failed-call charges have different accounting scopes. The reported results therefore do not establish an end-to-end token-efficiency or monetary-cost advantage.

Persistent provenance. The archived results snapshot retains briefs, point judgments, simulation records, retry attempts, policy-selection records, and the independent coordinator-retrieval pilot. Earlier attempts remain available with their provenance. Verification after restoring the mount archives to local scratch storage checked object hashes, reference integrity, baseline identity, and expected counts. The snapshot is intended for results review rather than full cache resumption. It omits duplicate long requests, redundant caches and logs, credentials, live databases, and model downloads.

F.3. Historical Optimization Details

Historical policy-selection records. The following records document selection of the frozen policies. The search was not rerun; current performance is evaluated using freshly rebuilt briefs in Section 5.3.

GroupBrief (optimized) policies. Both searches completed five rounds. The selected policies were `constraint_followups` from round 5 for `gpt-5.4-mini` and `rationale_followthrough` from round 4 for `gpt-5.6-luna`. In the historical evaluation after policy freezing, Test coverage increased on six of seven projects for `gpt-5.4-mini` and on all seven for `gpt-5.6-luna`, with no judge errors. These projects were excluded from this search, but Test had been inspected in earlier experiments, making the transfer evidence exploratory.

Both GroupBrief (optimized) variants retain GroupBrief's tools, step and observation budgets, and final-synthesis structure while revising retrieval and synthesis instructions. `gpt-5.4-mini` follows the central choice in the request to its thread, reasons, boundaries, prerequisites, and later pre-cutoff mentions, while keeping the proposer, reviewer, and eventual user distinct. Its synthesis connects each supported intended benefit to the object, affected role, and failure it aims to prevent. `gpt-5.6-luna` follows each work item's thread when a proposal lacks its rationale, then inventories evidence by item and speaker and states safeguard-to-purpose links separately. Both increased coverage in all four reference categories on the historical Valid and Test samples (Table 5).

These comparisons do not isolate the source of the gains or establish an advantage of synthesis optimization over retrieval optimization.

Category scores and search trajectory. Table 5 reports GroupBrief (optimized) for the two writers, whose implementation identifiers are `constraint_followups` and `rationale_followthrough`. Team process and emergent state are separated here and combined as past state elsewhere. Overall coverage is weighted by point, and gains are computed from unrounded scores.

Table 5 | Historical search-sample category coverage, retained only as policy-selection provenance. Current rebuilt category scores appear in Figure 6. Scores are percentages; gains are pp.

		Valid			Test		
Writer	Category	GroupBrief	GroupBrief (optimized)	Gain	GroupBrief	GroupBrief (optimized)	Gain
gpt-5.4-mini	Team process	40.31	50.42	+10.11	40.58	46.56	+5.99
	Team emergent state	25.74	33.82	+8.09	29.89	38.59	+8.70
	Future friction	12.18	18.50	+6.32	18.09	20.45	+2.36
	Future opportunities	20.34	34.00	+13.67	27.27	32.97	+5.70
gpt-5.6-luna	Team process	52.67	61.80	+9.13	55.43	58.31	+2.88
	Team emergent state	38.97	60.29	+21.32	36.41	50.00	+13.59
	Future friction	22.83	30.68	+7.85	23.25	33.13	+9.88
	Future opportunities	32.31	47.78	+15.47	35.82	48.24	+12.42

The incumbent Valid trajectory for `gpt-5.4-mini` is 26.95, 32.37, 32.37, 34.84, 36.60, and 37.02% from GroupBrief through rounds 1–5; `gpt-5.6-luna`'s is 38.80, 44.70, 49.70, 49.79, 49.88, and 49.88%. The incumbent remains unchanged in some rounds. `gpt-5.4-mini`'s final policy scores lower on Train than the preceding `workflow_effects` (35.77% versus 36.20%) but higher on Valid (37.02% versus 36.60%), illustrating why selection uses Valid. A `consequence_ledger` candidate has an incomplete evaluation with five judge errors and is not treated as a valid score. Both optimized writers are GPT-family models, and there is no randomized budget-matched mechanism ablation.

F.3.1. Complete Candidate Search

Tables 6 and 7 show how each historical search selected its incumbent, retaining both proposals from every round. Train and Valid use fixed splits of 828 and 234 requests, respectively. "Selected" marks a candidate that became the incumbent at the end of its round; "Retained" means the preceding incumbent remained. An unselected candidate can still outperform GroupBrief. The declared axis records the proposer's edit category, not a causal attribution of the resulting score.

Table 6 | All ten candidates proposed for `gpt-5.4-mini`. Coverage scores are percentages. "Incomplete" denotes five Train judge errors; "Not evaluated" denotes the absence of a Valid evaluation. In round 2, the retained incumbent is `causal_claims`.

Round	Candidate	Declared axis	Train	Valid	Round outcome
1	`anchored_followups`	Search	30.40	30.70	Not selected
1	`causal_claims`	Synthesis	35.13	32.37	Selected
2	`consequence_ledger`	Synthesis	Incomplete	Not evaluated	Ineligible
2	`rationale_threads`	Search	32.02	32.34	Retained
3	`object_paired_probes`	Search	35.46	34.69	Not selected
3	`recipient_effects`	Synthesis	34.43	34.84	Selected
4	`paired_role_checks`	Search	35.72	36.30	Not selected
4	`workflow_effects`	Synthesis	36.20	36.60	Selected
5	`conditional_effects`	Synthesis	34.07	34.10	Not selected
5	`constraint_followups`	Search	35.77	37.02	Selected

Eligibility and missing evaluations. The anti-regression tolerance is $\epsilon = 0.02$, or two percentage points, relative to GroupBrief on each split. A complete candidate must pass the Train gate before

Table 7 | All ten candidates proposed for gpt-5.6-luna. Coverage scores are percentages. artifact_boundaries completes Train but does not qualify for Valid. In round 5, the retained incumbent is rationale_followthrough.

Round	Candidate	Declared axis	Train	Valid	Round outcome
1	causal_lookahead	Synthesis	45.66	44.70	Selected
1	anchored_followup	Search	38.32	39.46	Not selected
2	artifact_boundaries	Synthesis	36.24	Not evaluated	Below Train gate
2	causal_inventory	Synthesis	46.39	49.70	Selected
3	object_purpose_ledger	Synthesis	42.66	45.21	Not selected
3	anchored_workstreams	Search	47.42	49.79	Selected
4	rationale_followthrough	Search	48.46	49.88	Selected
4	request_first_claims	Synthesis	39.74	37.16	Not selected
5	paired_boundary_probe	Search	47.94	49.82	Retained
5	artifact_effect_pairs	Synthesis	46.93	48.57	Not selected

receiving a Valid evaluation. gpt-5.4-mini's consequence_ledger fails the completion requirement because of five judge errors. gpt-5.6-luna's artifact_boundaries completes Train at 36.24%, below its 36.32% eligibility floor (38.32% minus two points). These are distinct exclusion reasons; neither is assigned a zero Valid score. gpt-5.6-luna's request_first_claims, by contrast, passes Train and is evaluated on Valid. Its 37.16% Valid coverage exceeds the 36.80% eligibility floor but remains below both GroupBrief and incumbent. Across both searches, 20 proposals undergo Train evaluation, of which 19 complete; 18 receive Valid evaluation, in addition to the two GroupBrief baselines. The tolerance is neither a required improvement nor a significance threshold.

Selection reversals and diminishing increments. Validation feedback can reverse the ordering suggested by Train. In gpt-5.4-mini's third round, object_paired_probes exceeds recipient_effects on Train (35.46% versus 34.43%) but trails it on Valid (34.69% versus 34.84%). For gpt-5.6-luna, most improvement arrives early: the first two selected candidates contribute 10.90 of the total 11.08 pp Valid gain. Rounds 3 and 4 each add 0.09 pp, and round 5 retains rationale_followthrough because the best new candidate, paired_boundary_probe, reaches 49.82% versus the incumbent's 49.88%. Selection uses unrounded scores. These increments characterize a feedback-dependent search, rather than a controlled retrieval or synthesis ablation; selection alone does not establish the effectiveness of a proposed mechanism.

F.4. GroupBrief (optimized) versus Baselines

Table 8 evaluates the current rebuilt briefs on identical Test points, separately from the historical search samples. The primary GroupBrief (optimized) sample with gpt-5.4-mini is the reference; regenerated GroupBrief and GroupBrief (optimized) briefs form separate samples. These descriptive comparisons fix the writer while retaining unequal retrieval and memory-construction budgets.

F.5. Completed Test-Only Simulation

Table 9 compares completed simulation results on the 289 Test tasks using replicate 0. Methods share matched labels, initial-state inputs, and the same selected no-brief baseline, while each preference rate is conditional on its displayed valid denominator. Train/Valid outcomes, unavailable-scenario placeholders, and historical two-seed aggregates are excluded.

Independent coordinator-retrieval ablation. The pilot uses two hash-selected tasks from each Train project, giving forty tasks across twenty projects. Both conditions generate fresh trajectories and condition-matched no-brief controls from identical original briefs and budgets. The four methods yield 320 comparisons and 400 trajectories. With retrieval enabled, there are 135 stable comparisons, 21 disagreements, and four simulation failures; with retrieval disabled, the

Table 8 | Test coverage (%) with gpt-5.4-mini on 289 requests and 2,162 points. Differences are row minus primary GroupBrief (optimized) coverage, in pp.

Method	Test	Δ Test
GroupBrief (optimized) with gpt-5.4-mini	35.36	+0.00
GroupBrief with gpt-5.4-mini	32.82	-2.54
Hindsight	33.02	-2.34
ReadAgent-S	27.57	-7.79
Full context	24.31	-11.05
BM25	20.58	-14.78
Contriever	17.90	-17.46
DRAGON+	18.41	-16.95
A-MEM	19.82	-15.54
HippoRAG 2	17.69	-17.67
Mem0	14.43	-20.93
GroupBrief with gpt-5.4-mini (regenerated)	34.14	-1.23
GroupBrief (optimized) with gpt-5.4-mini (regenerated)	36.96	+1.60

Table 9 | Completed Test simulation results (%) under the uniform bounded retry policy. Every method has 289 processed tasks. Rates from different valid sets must not be subtracted as paired effects.

Method	Valid	Preference
GroupBrief with gpt-5.4-mini	238	62.18
Hindsight	244	64.75
ReadAgent-S	249	64.06
Full context	244	60.25
BM25	256	59.57
Contriever	245	64.08
DRAGON+	252	60.12
A-MEM	242	62.19
HippoRAG 2	247	63.56
Mem0	256	62.11
GroupBrief (optimized) with gpt-5.4-mini	242	64.46
GroupBrief with gpt-5.6-sol	232	68.53
GroupBrief with qwen3.8-27b	236	62.71
GroupBrief with gpt-5.5	235	64.89
GroupBrief with gpt-6-astra	239	64.85
GroupBrief with gpt-5.6-luna	254	65.55
GroupBrief with gpt-6-luna	240	66.88
GroupBrief with gpt-6-sol	238	58.40
GroupBrief with qwen3.5-9b	238	57.77
GroupBrief (optimized) with gpt-5.6-luna	247	68.42
GroupBrief with gpt-5.4-mini (regenerated)	242	64.46
GroupBrief (optimized) with gpt-5.4-mini (regenerated)	253	62.25

corresponding counts are 133, 24, and three. All eighty no-brief trajectories complete. There are no judge failures, and trajectories with retrieval disabled execute zero coordinator search/read actions.

Table 10 | Train pilot, forty planned pairs per cell. Each condition uses its own fresh no-brief control; the marginal rates are not a fixed-population condition effect.

Method	Valid enabled	Preference	Valid disabled	Preference
GroupBrief with gpt-5.4-mini	34	50.00	35	58.57
GroupBrief (optimized) with gpt-5.4-mini	33	60.61	33	63.64
Hindsight	34	58.82	34	58.82
BM25	34	47.06	31	64.52

On the same 22 tasks valid for both GroupBrief and GroupBrief (optimized) in both conditions, the GroupBrief (optimized) minus GroupBrief gap is +9.09 pp with retrieval and +2.27 pp without it. The gap change is -6.82 pp. The no-brief condition used retrieval in only three of forty control tasks. The pilot is separate from the completed main evaluation and does not establish greater method separation or a detrimental effect of retrieval.